

MTH166

Lecture-10

**Solution of 2nd Order Homogeneous LDE
with Constant Coefficients-II**

Topic:

Solution of 2nd order Homogeneous LDE with Constant coefficients

Learning Outcomes:

1. Formulation of 2nd order homogeneous LDE when roots are given.
2. Solution of Initial Value Problems (IVP) and Boundary Value Problems (BVP).

Formulation of LDE of the form: $ay'' + by' + cy = 0$ when Roots are given:

Let the two given roots be: m_1 and m_2 .

Then required 2nd order homogeneous LDE is:

$$y'' - (\text{sum of roots}) y' + (\text{Product of roots}) y = 0$$

$$\text{i.e. } y'' - (m_1 + m_2)y' + (m_1m_2)y = 0$$

or

Then required 2nd order homogeneous LDE is:

$$(D - m_1)(D - m_2)y = 0 \text{ where } D \equiv \frac{d}{dx}$$

Find a LDE of the form: $ay'' + by' + cy = 0$ for which the following functions are solutions:

Problem 1. (e^{3x}, e^{-2x})

Solution: Comparing with: (e^{m_1x}, e^{m_2x})

We have: $m_1 = 3, m_2 = -2$

Then required 2nd order homogeneous LDE is:

$$y'' - (\text{sum of roots})y' + (\text{Product of roots})y = 0$$

$$\text{i.e. } y'' - (m_1 + m_2)y' + (m_1m_2)y = 0$$

$$\Rightarrow y'' - (3 - 2)y' + (3)(-2)y = 0$$

$$\Rightarrow y'' - y' - 6y = 0 \quad \textbf{Answer.}$$

Problem 2. $(1, e^{-2x})$

Solution: Here $(1, e^{-2x}) = (e^{0x}, e^{-2x})$

Comparing with: (e^{m_1x}, e^{m_2x})

We have: $m_1 = 0, m_2 = -2$

Then required 2nd order homogeneous LDE is:

$$y'' - (\text{sum of roots}) y' + (\text{Product of roots})y = 0$$

$$\text{i.e. } y'' - (m_1 + m_2)y' + (m_1m_2)y = 0$$

$$\Rightarrow y'' - (0 - 2)y' + (0)(-2)y = 0$$

$$\Rightarrow y'' + 2y' = 0 \quad \textbf{Answer.}$$

Problem 3. (e^{2x}, xe^{2x})

Solution: Comparing with: (e^{m_1x}, xe^{m_2x})

We have: $m_1 = 2, m_2 = 2$

Then required 2nd order homogeneous LDE is:

$$y'' - (\text{sum of roots}) y' + (\text{Product of roots})y = 0$$

$$\text{i.e. } y'' - (m_1 + m_2)y' + (m_1m_2)y = 0$$

$$\Rightarrow y'' - (2 + 2)y' + (2)(2)y = 0$$

$$\Rightarrow y'' - 4y' + 4y = 0 \quad \textbf{Answer.}$$

Problem 4. (e^{-3ix}, e^{3ix})

Solution: Comparing with: (e^{m_1x}, e^{m_2x})

We have: $m_1 = -3i, m_2 = 3i$

Then required 2nd order homogeneous LDE is:

$$y'' - (\text{sum of roots}) y' + (\text{Product of roots})y = 0$$

$$\text{i.e. } y'' - (m_1 + m_2)y' + (m_1m_2)y = 0$$

$$\Rightarrow y'' - (-3i + 3i)y' + (-3i)(3i)y = 0$$

$$\Rightarrow y'' + 9y = 0 \quad \textbf{Answer.} \quad (i^2 = -1)$$

Problem 5. $(e^{(5+3i)x}, e^{(5-3i)x})$

Solution: Comparing with: (e^{m_1x}, e^{m_2x})

We have: $m_1 = 5 + 3i, m_2 = 5 - 3i$

Then required 2nd order homogeneous LDE is:

$$y'' - (\text{sum of roots}) y' + (\text{Product of roots})y = 0$$

$$\text{i.e. } y'' - (m_1 + m_2)y' + (m_1m_2)y = 0$$

$$\Rightarrow y'' - [(5 + 3i) + (5 - 3i)]y' + (5 + 3i)(5 - 3i)y = 0$$

$$\Rightarrow y'' - 10y' + 34y = 0 \quad \textbf{Answer.} \quad (i^2 = -1)$$

Polling Question

If (e^{-3x}, e^{2x}) are the roots, then the corresponding LDE is:

(A) $y'' + y' + 6y = 0$

(B) $y'' + y' - 6y = 0$

(C) $y'' - y' - 6y = 0$

Problem: Solve the Initial value problem: $y'' - y = 0$, $y(0) = 0$, $y'(0) = 2$.

Solution: The given equation is:

$$y'' - y = 0 \quad (1)$$

Such that: $y(0) = 0$, $y'(0) = 2$

S.F. : $(D^2 - 1)y = 0$ where $D \equiv \frac{d}{dx}$

A.E. : $(D^2 - 1) = 0 \Rightarrow D^2 = 1 \Rightarrow D = \pm 1$ (real and unequal roots)

Let $m_1 = 1$ and $m_2 = -1$

$\therefore$ General Solution of equation (1) is given by:

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$$

$$\Rightarrow y = c_1 e^{1x} + c_2 e^{-1x}$$

$$\Rightarrow y(x) = c_1 e^x + c_2 e^{-x} \quad (2)$$

$$\Rightarrow y'(x) = c_1 e^x - c_2 e^{-x} \quad (3)$$

Using $y(0) = 0$ in equation (2), we get:

$$y(0) = c_1 e^0 + c_2 e^{-0} \quad \Rightarrow 0 = c_1 + c_2 \quad (4)$$

Using $y'(0) = 2$ in equation (3), we get:

$$y'(0) = c_1 e^0 - c_2 e^{-0} \quad \Rightarrow 2 = c_1 - c_2 \quad (5)$$

Solving equations (4) and (5), we get: $c_1 = 1, c_2 = -1$

Putting these values of c_1 and c_2 in equation (2), we get:

$$y(x) = e^x - e^{-x} \quad \textbf{Answer.}$$

Problem: Solve the Boundary value problem: $y'' - 4y' + 3y = 0$ such that $y(0) = 1, y(1) = 0$.

Solution: The given equation is:

$$y'' - 4y' + 3y = 0 \quad (1)$$

Such that: $y(0) = 1, y(1) = 0$

S.F. : $(D^2 - 4D + 3)y = 0$ where $D \equiv \frac{d}{dx}$

A.E. : $(D^2 - 4D + 3) = 0 \Rightarrow (D - 1)(D - 3) = 0$

$\Rightarrow D = 1, 3$ (real and unequal roots)

Let $m_1 = 1$ and $m_2 = 3$

$\therefore$ General Solution of equation (1) is given by:

$$y = c_1 e^{m_1 x} + c_2 e^{m_2 x}$$

$$\Rightarrow y = c_1 e^{1x} + c_2 e^{3x}$$

$$y(x) = c_1 e^{1x} + c_2 e^{3x} \quad (2)$$

Using $y(0) = 1$, we get:

$$y(0) = c_1 e^0 + c_2 e^0 \quad \Rightarrow 1 = c_1 + c_2 \quad (3)$$

Using $y(1) = 0$, we get:

$$y(1) = c_1 e^1 + c_2 e^3 \quad \Rightarrow 0 = c_1 e^1 + c_2 e^3 \quad (4)$$

Solving equations (3) and (4), we get: $c_1 = \frac{e^2}{e^2-1}$ and $c_2 = \frac{1}{e^2-1}$

Putting these values of c_1 and c_2 in equation (2), we get:

$$y(x) = \frac{e^2}{e^2-1} e^x + \frac{1}{e^2-1} e^{3x} \quad \textbf{Answer.}$$

Polling Question

A Linear differential equation with conditions given as:

$y(a) = 0$ and $y(b) = 1$ (say) is called:

(A) Initial value problem

(B) Boundary value problem

